Electron microscope

Electron microscopes

- Electron microscopes are far more powerful than light microscopes.
- They were invented in the 1930s.
- Electron microscopes use electrons that are sped up in a vacuum.
- Beams of electrons are directed at the specimen.
- Electrons are either absorbed or scattered to form an image on an electron-sensitive photographic plate.
- Electron microscopes are large, heavy and expensive.
- The microscope magnifies microorganisms, cells, crystals, metals and large molecules.

Read the information above and then answer the following questions.

☞ **1** In what way could electron microscopes be useful in medical research?

_____

_____

2 Living cells can't survive in a high vacuum. How is this a problem when studying cells using an electron microscope? _____

_____

3 List reasons why schools don't have electron microscopes. _____

_____

4 How might palaeontologists (scientists who study fossils) find this microscope useful?

_____

_____

© Beth Montgomery and Blake Education 2012
© Blake Education 978 1 92571 040 3 Achieve! Science Understanding Biology Yr 7 - 8